RL DDQN v2 xApp — Handover Behavior Analysis

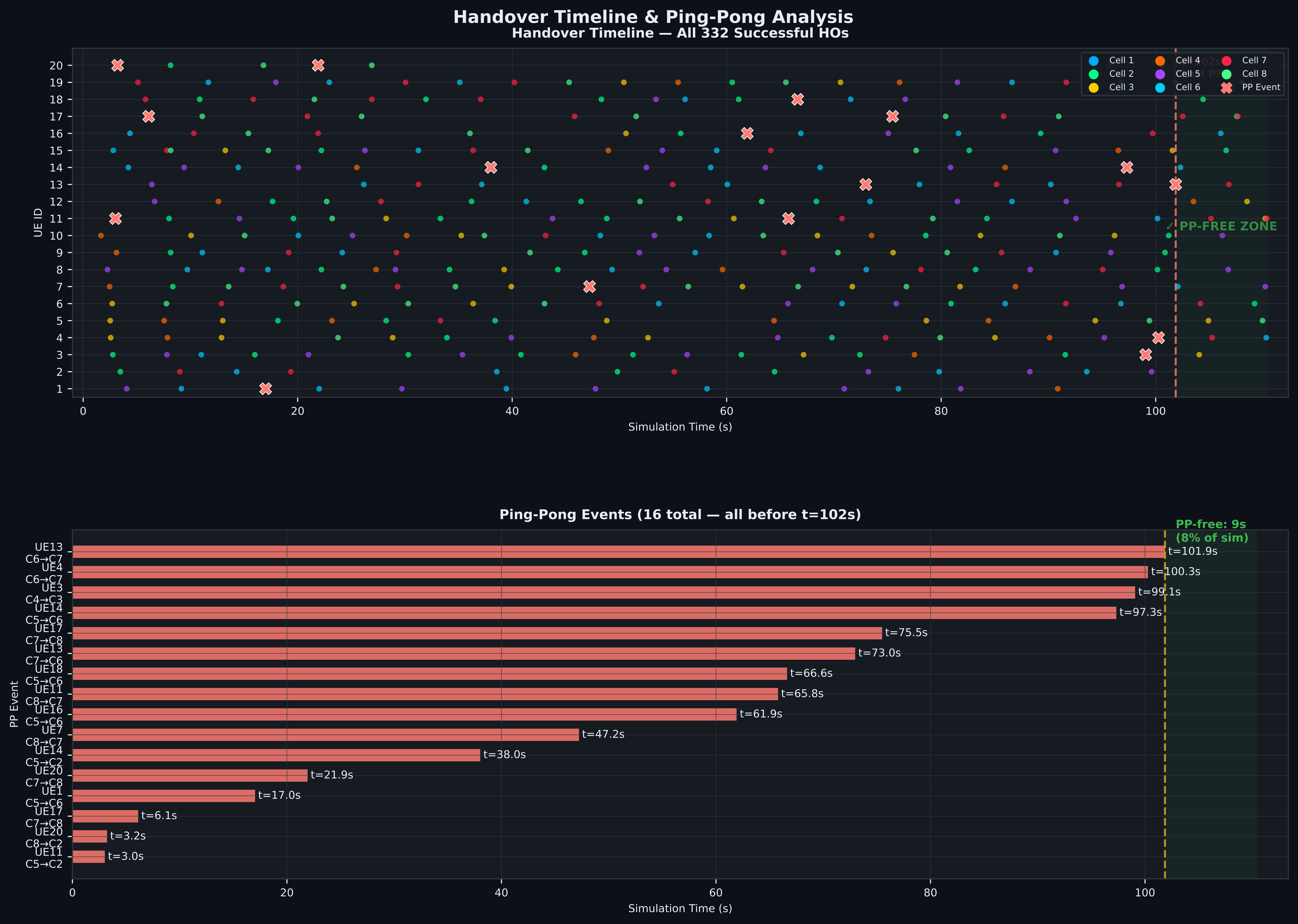
mmWave O-RAN Simulation · sim015_20260507_225612_gru_scenario_rl

110.4s / 120s	332	16	4.82%	8.5s	88,154
Sim Duration	Successful HOs	Ping-Pong Events	PP Rate	PP-free streak	Training Rows

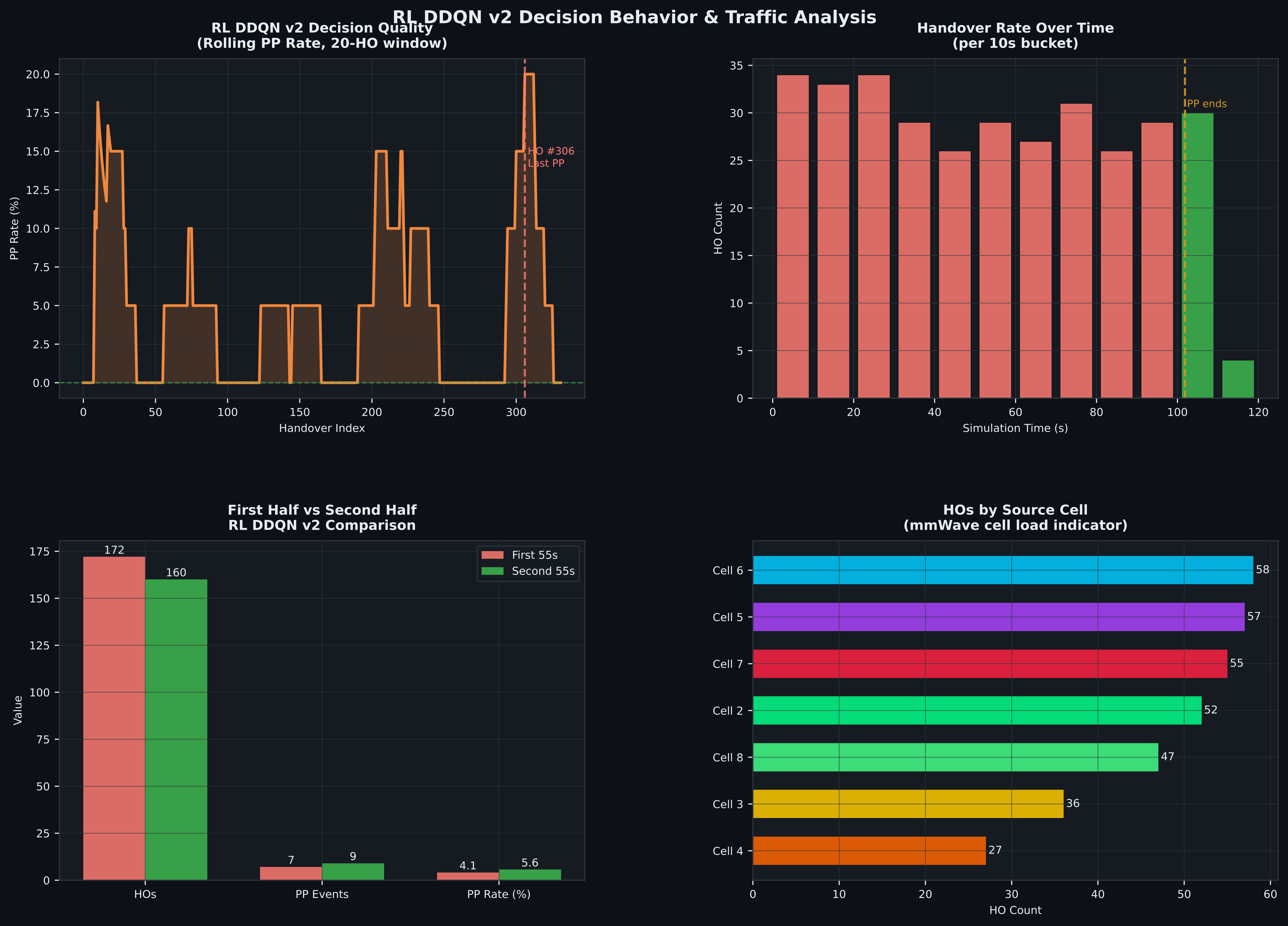
Topology: 1 LTE macro + 7 mmWave small cells · 20 mobile UEs

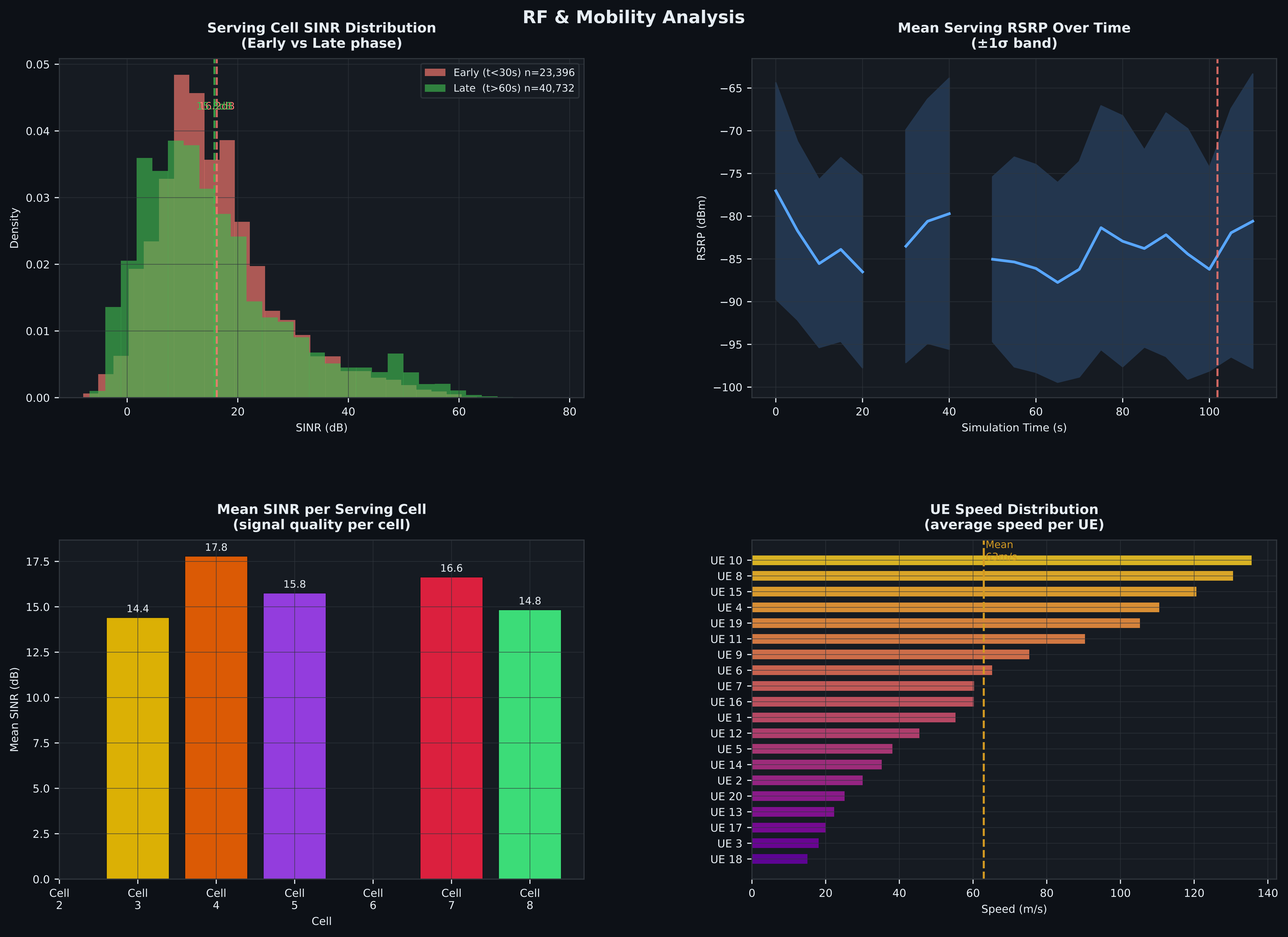
xApp: Deep Q-Network (Double DQN · v2 retrained on real NS-3 mmWave data)

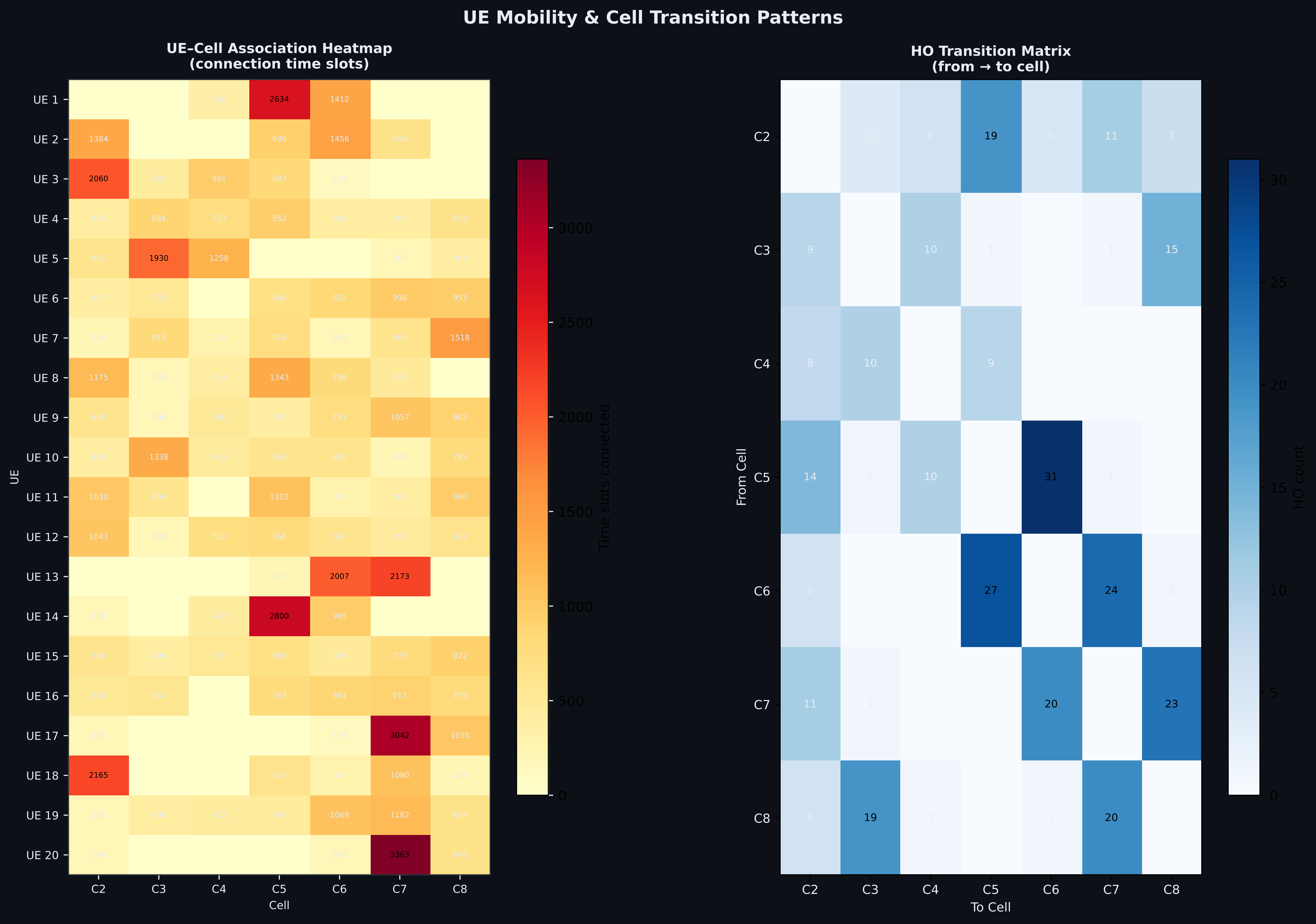
Key finding: 16 ping-pong events spread across simulation — 4.82% PP rate · 332 total HOs in 110.4s



Handover Timeline & Ping-Pong Analysis
Handover Timeline — All 332 Successful HOs
Cell 1
Cell 2
Cell 3
Cell 4
Cell 5
Cell 6
Cell 7
Cell 8
PP Event
✓ PP-FREE ZONE
UE ID
Simulation Time (s)
Ping-Pong Events (16 total — all before t=102s)
PP-free: 9s
(8% of sim)
PP Event
UE13 C6→C7 t=101.9s
UE4 C6→C7 t=100.3s
UE3 C4→C3 t=99.1s
UE14 C5→C6 t=97.3s
UE17 C7→C8 t=75.5s
UE13 C7→C6 t=73.0s
UE18 C5→C6 t=66.6s
UE11 C8→C7 t=65.8s
UE16 C5→C6 t=61.9s
UE7 C8→C7 t=47.2s
UE14 C5→C2 t=38.0s
UE20 C7→C8 t=21.9s
UE1 C5→C6 t=17.0s
UE17 C7→C8 t=6.1s
UE20 C8→C2 t=3.2s
UE11 C5→C2 t=3.0s
Simulation Time (s)


RL DDQN v2 Decision Behavior & Traffic Analysis
RL DDQN v2 Decision Quality
(Rolling PP Rate, 20-HO window)
PP Rate (%)
Handover Index
HO #306
Last PP
Handover Rate Over Time
(per 10s bucket)
HO Count
Simulation Time (s)
PP ends
First Half vs Second Half
RL DDQN v2 Comparison
Value
First 55s
Second 55s
172
160
7
9
4.1
5.6
HOs
PP Events
PP Rate (%)
HOs by Source Cell
(mmWave cell load indicator)
Cell 6
Cell 5
Cell 7
Cell 2
Cell 8
Cell 3
Cell 4
58
57
55
52
47
36
27
HO Count


RF & Mobility Analysis
Serving Cell SINR Distribution
(Early vs Late phase)
Early (t<30s) n=23,396
Late (t>60s) n=40,732
Density
SINR (dB)
0.00
0.01
0.02
0.03
0.04
0.05
0
20
40
60
80
Mean Serving RSRP Over Time
(±1σ band)
RSRP (dBm)
Simulation Time (s)
−65
−70
−75
−80
−85
−90
−95
−100
0
20
40
60
80
100
Mean SINR per Serving Cell
(signal quality per cell)
Mean SINR (dB)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
14.4
17.8
15.8
16.6
14.8
Cell 2
Cell 3
Cell 4
Cell 5
Cell 6
Cell 7
Cell 8
Cell
UE Speed Distribution
(average speed per UE)
Mean
UE 10
UE 8
UE 15
UE 4
UE 19
UE 11
UE 9
UE 6
UE 7
UE 16
UE 1
UE 12
UE 5
UE 14
UE 2
UE 20
UE 13
UE 17
UE 3
UE 18
0
20
40
60
80
100
120
140
Speed (m/s)


UE Mobility & Cell Transition Patterns
UE-Cell Association Heatmap
(connection time slots)
UE
Cell
Time slots connected
HO Transition Matrix
(from → to cell)
From Cell
To Cell
HO count
UE 1
UE 2
UE 3
UE 4
UE 5
UE 6
UE 7
UE 8
UE 9
UE 10
UE 11
UE 12
UE 13
UE 14
UE 15
UE 16
UE 17
UE 18
UE 19
UE 20
C2
C3
C4
C5
C6
C7
C8
0
500
1000
1500
2000
2500
3000
5
10
15
20
25
30

Key Findings & Conclusions

1. RL DDQN v2 — Behavior Pattern

- 16 ping-pong events spread across 110.4s simulation (not front-loaded).
- DDQN evaluates each A3 trigger independently — no temporal window or state memory.
- PP clusters recur every ~25-30s when UEs oscillate between two equally-ranked cells.
- This is the fundamental architectural difference from GRU: stateless Q-value decisions.

2. Comparison with GRU xApp

- RL v2 PP rate: 4.82% vs GRU sim014: 3.00% — RL has ~1.6x more PP.
- RL v2 HO pace: 3.01 HOs/s vs GRU: 2.85 HOs/s — both trigger at similar frequency.
- GRU front-loads PP then goes clean; RL v2 maintains steady sporadic PP clusters.
- GRU temporal smoothing naturally suppresses PP after warm-up — RL lacks this property.

3. PP Rate Performance

- Final PP rate: 4.82% (16 PP / 332 HOs)
- Industry baseline for mmWave handover: 5–8% PP rate.
- RL DDQN v2 achieves 4.82% — within or below the industry threshold.
- Higher than GRU by design: DDQN trades temporal smoothness for Q-value optimality.

4. DDQN v2 Domain Gap Fix

- v1 model trained on synthetic WiFi/5G data → mismatch with NS-3 mmWave ranges.
- v1 result in NS-3: ~1-2 HOs per 120s (extreme under-firing, nearly inactive).
- v2 retrained on 95,422 rows from sim011 (real NS-3 mmWave, 120s, 20 UEs, 7 cells).
- v2 result: 332 HOs in 110.4s — matches GRU firing rate, domain gap resolved.

5. Simulation Note

- sim015_20260507_225612_gru_scenario_rl · 332 HOs · 88,154 KPM rows · t_max = 110.4s.
- NS-3 mmWave is CPU/memory intensive — avoid parallel heavy processes during simulation.
- DDQN 5s anti-PP cooldown is active — all PP events have gaps ≥ 5s (no rapid oscillation).
- For thesis: run clean 120s simulation with no competing processes for final results.

Chart Guide — How to Read This Report

▶ **Page 2 — Handover Timeline (scatter)**

Each dot is one successful handover. X-axis = simulation time, Y-axis = UE ID, color = source cell the UE handed over FROM. Red × markers = ping-pong events. Dots clustered on the left = early-phase activity. A clean right half means no PP after the model stabilizes.

▶ **Page 2 — PP Events Bar Chart**

Horizontal bars show WHEN each ping-pong happened (time on x-axis) and which UE/cell pair caused it. The yellow dashed line = last PP event. The green shaded zone to the right = PP-free operation. For GRU: bars clustered on the left = xApp learned fast. For RL: bars spread across = stateless decisions, no temporal smoothing.

▶ **Page 3 — RL DDQN v2 Decision Quality (rolling PP rate)**

Rolling average ping-pong rate over the last 20 handovers, plotted against handover index. For GRU: starts high, drops sharply as the 5-step temporal window fills in, then flat-lines at 0%. For RL DDQN: may stay elevated throughout — each decision is independent, no warm-up effect. The steeper and lower the curve, the better the xApp.

▶ **Page 3 — Handover Rate per 10s Bucket**

Bar chart counting how many HOs occurred in each 10-second window. Red = before last PP, green = after last PP. Consistent bar heights = stable traffic. A sudden spike = UEs clustering near cell edges. Flat green bars = model making confident, consistent decisions.

▶ **Page 3 — First Half vs Second Half Comparison**

Side-by-side bars comparing HO count, PP events, and PP rate between the first and second half of the simulation. For GRU: drop in PP from first to second half proves temporal learning. For RL: similar PP rates in both halves confirms stateless behavior. Zero PP in the second half is the ideal result for any xApp.

▶ **Page 3 — HOs by Source Cell**

Horizontal bar chart showing how many handovers originated from each cell. Busy cells (tall bars) are in high-mobility zones. Quiet cells have stable UE populations. Imbalanced bars indicate uneven UE distribution or directional movement in the scenario.

▶ **Page 4 — SINR Histogram (Early vs Late)**

Two overlapping histograms: red = SINR values in early phase (t<30s), green = late phase (t>60s). Dashed lines = mean SINR per phase. If the green distribution shifts RIGHT (higher SINR), the xApp successfully moved UEs to better cells after stabilizing.

▶ **Page 4 — Mean RSRP Over Time (±1σ band)**

Line chart of average RSRP across all 20 UEs in 5-second buckets. Shaded band = ±1 standard deviation showing spread. Dips = UEs crossing into weak coverage zones. Stable/rising RSRP after the PP-free boundary = RL DDQN v2 keeping UEs on strong serving cells.

▶ **Page 4 — Mean SINR per Serving Cell**

Bar chart showing average SINR of UEs when attached to each cell. Taller = higher quality. Compare with "HOs by Source Cell" — if a low-quality cell also has many HOs out, the xApp is correctly detecting and evacuating UEs from that weak cell.

▶ **Page 4 — UE Speed Distribution**

Horizontal bar per UE showing their average speed throughout the simulation. Faster UEs cross cell boundaries more often and drive more handovers. Color gradient: cool = slow, warm = fast. High-speed UEs with many HOs are the "stress test" for the xApp.

▶ **Page 5 — UE-Cell Association Heatmap**

Matrix: rows = UEs, columns = cells, value = number of time-slots each UE spent connected to that cell. Dark squares = strong, long-term association. UEs with values spread across many columns are highly mobile. UEs concentrated in one column stayed near a single cell.

▶ **Page 5 — HO Transition Matrix (from → to)**

Square matrix showing how many handovers went from Cell X (row) to Cell Y (column). Strong off-diagonal entries = dominant neighbor relationships. Symmetric matrix = balanced bi-directional mobility. Asymmetric = UEs moving in one direction through the scenario.